

任课教师: _____

学号: _____

姓名: _____

班级: _____

装订线

装订线

装订线

西 安 电 子 科 技 大 学

考试时间 120 分钟

试 题

题号	一	二	三	四	五	六	七	总分
分数								

1. 考试形式: 闭卷 ☒ 开卷 ☐

2. 考试日期: 年 月 日 (答题内容请写在装订线外)

Problem 1. Analysis of algorithms (15 points.)

Fill in the table to give the corresponding code framework of the given order of growth.

Order of growth	Code framework
N	<pre>for (int i = 0; i < N; i++) { ... }</pre>
NlgN	
N²	

Problem 2. Union-Find. (10 points.)

Given the input pair sequence 0-6 6-1 4-5 6-4 2-4 6-3, draw the forest of trees represented by the `id[]` array after each input pair is processed until '0' and '5' are connected, by using *quick-union* and *weighted quick-union* respectively.

1. 每一步处理后的 `id[]` / `parent[]` ;
2. 对应的森林结构, 也就是树长什么样。
顺次执行, 知道0和5union后停机

Problem 3. Sorting. (20 points. Two parts, 10 points for each part)

(1) Fill in the blanks to complete the implementation of merge operation in *Mergesort*.

```
void merge(Comparable[] a, Comparable[] aux, int lo, int mid, int hi)
{
    assert isSorted(a, lo, mid);
    assert isSorted(a, mid+1, hi);
    for (int k = lo; k <= hi; k++)
        aux[k] = a[k];
    int i = lo, j = mid+1;
    for (int k = lo; k <= hi; k++)
    {
        if (i > mid) _____;
        else if (j > hi) _____;
        else if (less(aux[j], aux[i])) _____;
        else _____;
    }
    assert isSorted(a, lo, hi);
}
```

less (a, b)
一定是a<b的时候返回true

易错：
充分考虑else的情况：
包含了a[i]更小的
情况

(2) Show the trace of the partition() in *Quicksort* working on array E D S Y F U E S.

trace 一般就是让你把算法执行过程“追踪写出来”。
这题要的是一次 partition 的过程，不是完整 quicksort；重点写 i、j 怎么移动、每次交换后的数组，以及 pivot 最终位置。

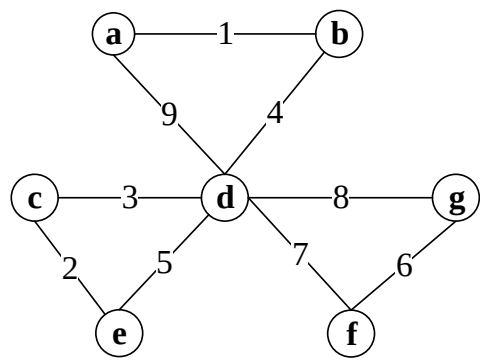
Problem 4. Binary search trees (BST). (10 points)

Consider the keys **R E S P O N D** that, draw the best-case BST and the worst-case BST.

最好情况就是尽量平衡（树高最小）（尽量平衡，让左右子树规模尽可能接近。）。
7 个节点刚好可以构造一棵满二叉树。
最坏情况就是完全退化成链表。

Problem 5. Minimum spanning trees (MST). (15 points. Two parts, 5 points for the first part and 10 points for the second part)

Given an edge-weighted graph G .

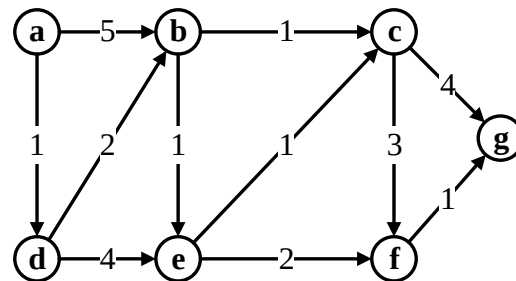


(1) Draw the MST of G .

(2) Give the process of using Kruskal’s algorithm to compute the MST of G .

Problem 6. Shortest paths. (15 points. Two parts, 10 points for the first part, 5 points for the second part)

Give an edge-weighted digraph as shown below.



(1) Fill in the table below after running Dijkstra's algorithm to compute the shortest path from vertex **a** to every other vertex.

vertex	distTo[]	edgeTo[]
a	0	null
b		
c		
d		
e		
f		
g		

(2) Give all occurred values of `distTo[e]` and `edgeTo[e]` when running Dijkstra's algorithm to compute the shortest path from vertex **a** to every other vertex.

<code>distTo[e]</code>	<code>edgeTo[e]</code>

Problem 7. Algorithm design. (15 points.)

Given an array containing N characters that range from 'a' to 'z', design an algorithm to preprocess the input in linear time and then answer any query about how many of the N characters are equal to or greater than β in constant time. Note that, β denotes a character in the range 'a' to 'z'. Your answer will be graded on correctness, efficiency, clarity, and conciseness.

附：单词释义表

题目	单词释义
1	order of growth 增长量级;就是Big-Theta
2	sequence 序列; respectively 分别地;
3	implementation 实现;
6	digraph 有向图; vertex 顶点; occurred 出现过的;
7	preprocess 预处理; query 查询; correctness, efficiency, clarity and conciseness 正确性、效率、清晰度和简洁性;